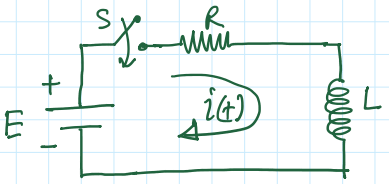
Lecture 9: Transient Analysis of R-L Series Circuit

In the previous class:



$$Ri + L\frac{di}{dt} = E$$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

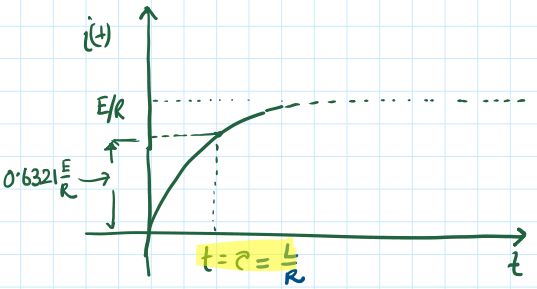
Which is of form: $\frac{di}{dt} + ai = b$, which has a solution:

$$i(t) = ke^{-at} + e^{-at}\int b\, e^{at}\, dt$$

$$i(t) = ke^{-at} + \frac{b}{a}$$

For this case: $a = \frac{R}{L}$, $b = \frac{E}{L}$, $\frac{b}{a} = \frac{E}{R}$

$$i(t) = ke^{-R/L\,t} + \frac{E}{R}$$



$$i(t) = ke^{-\frac{R}{L}t} + \frac{E}{R} = i_{tr} + i_{ss}$$

The constant k is solved using the initial condition. say $i = 0$ at $t = 0$

$\Rightarrow k = -\frac{E}{R}$

$$i(t) = \frac{E}{R}\left(1 - e^{-R/L\,t}\right) = \frac{E}{R}\left(1 - e^{-t/\tau}\right);\ \tau = \frac{L}{R}$$

Then we defined term τ = Time constant = $\frac{L}{R}$ (a time duration for which response reaches 63.21%)

Generalized Expression: Consider at $t = t_0$, $i(t_0) = i_{L0}$

We have:

$$i(t) = ke^{-\frac{R}{L}t} + \frac{E}{R}$$

$$i(t_0) = ke^{-\frac{R}{L}t_0} + \frac{E}{R}$$

$$\Rightarrow k = \left(i_L(t_0) - \frac{E}{R}\right)e^{\frac{R}{L}t_0} = \left(i_{L0} - \frac{E}{R}\right)e^{R/L\,t_0}$$

$$\Rightarrow i(t) = \left(i_{L0} - \frac{E}{R}\right)e^{-\frac{R}{L}(t - t_0)} + \frac{E}{R}$$

Here we can denote $i_i = i_{L0}$

$i_f = E/R$

$$i(t) = i_f + (i_i - i_f)e^{-\frac{R}{L}(t - t_0)} \quad \text{------ (I)}$$

exponential drop to zero from initial value ($i_i - i_f$)

exponential drop to zero
from initial value $(i_i - i_f)$
This gives, $i(t_0) = i_i$
$i(\infty) = i_f$

The above equation can also be written as

$$i(t) = i_f + (i_i - i_f)\, e^{-\frac{R}{L}(t-t_0)}$$

$$+ i_i - i_i$$

$$\boxed{i(t) = i_i + (i_f - i_i)\left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\}} \quad ---- \quad (II)$$

exponential rise:
from 0 to $(i_f - i_i)$
Thus ensuring: $i(t_0) = i_i = i_{L0}$
$i(\infty) = i_f = E/R$

The 3rd form of above expression can be written as,

$$i(t) = i_i + i_f \left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\} - i_i \left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\}$$

$$i(t) = i_i e^{-\frac{R}{L}(t-t_0)} + i_f \left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\} \quad ---- \quad (III)$$

$i_i e^{-\frac{R}{L}(t-t_0)}$ ↓ drives $i_i \to 0$

$i_f \left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\}$ ↓ drives 0 to i_f

Thus $i(t_0) = i_i$
$i(\infty) = i_f$

These three forms are summerised below.

$t = 0;\ i_L = 0$	$t = t_0,\ i_L = i_{L0} = i_i$
$i(t) = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$	Form-I: $i(t) = i_i e^{-\frac{R}{L}(t-t_0)} + i_f \left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\}$ Form II $i(t) = i_i + (i_f - i_i)\left\{1 - e^{-\frac{R}{L}(t-t_0)}\right\}$ Form-III $i(t) = i_f + (i_i - i_f)\, e^{-\frac{R}{L}(t-t_0)}$

What about voltage across inductance v_L ?

$$v_L = L\frac{di_L}{dt} = L * \frac{d}{dt}\left[\frac{E}{R}\left(1-e^{-R/Lt}\right)\right]$$

$$= L * \frac{E}{R} * \left(-e^{-\frac{R}{L}t}\right)\left(-\frac{R}{L}\right)$$

$$v_L(t) = E e^{-\frac{R}{L}t}$$

$$v_L(0) = E$$

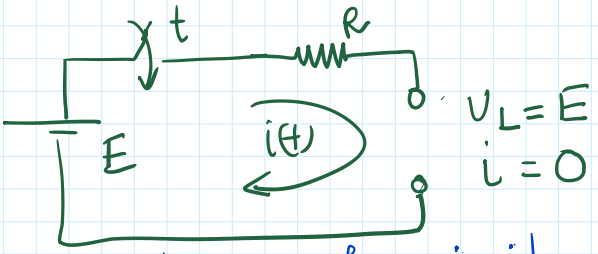
$$v_L(\infty) = 0$$

Note:

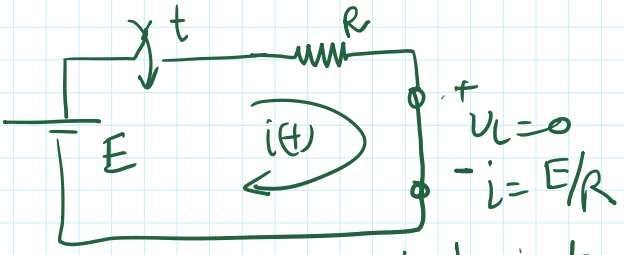
$$i_L(0) = 0$$

$$i_L(\infty) = E/R$$

	$t=0$	$t\to\infty$
i_L	0	E/R
v_L	E	0
Nature	open	short



L behaves as open circuit at $t=0$

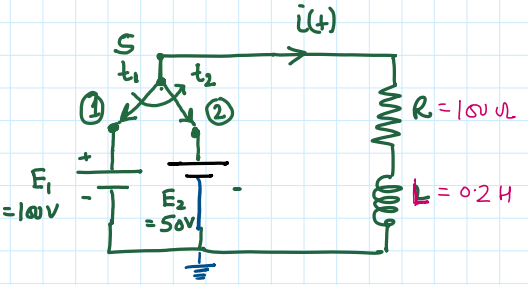


L behaves as short-circuit at $t\to\infty$

Example: In the network shown, switch S is closed on position 1 at $t=0$ and then opened to position 2 at $t = 500\ \mu sec$. Determine the equation for current, $i(t)$ when

a) For the shown direction of batteries

b) For reversed direction of 50V battery.



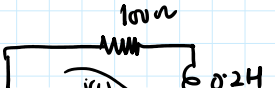
(a) At $t=0$; S is closed for position ①

At $t=0$, $i_L = 0$; $\frac{R}{L} = \frac{100}{0.2} = 500\ sec^{-1}$

$i_i = 0$, $i_f = \frac{E}{R} = \frac{100}{100} = 1A$

$$i(t) = i_f + (i_i - i_f)e^{-\frac{R}{L}t}$$

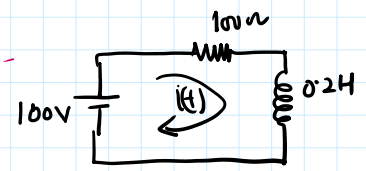
$$= 1 + (0-1)e^{-500\times10^{-6}t}$$



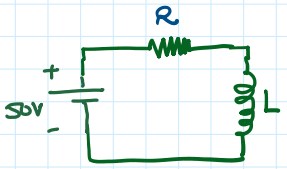
$i(t) = i_f + (i_i - i_f)\, e^{-500\times10^{-6}t}$

$= 1 + (0-1)\, e^{-500\times10^{-6}t}$

$i(t) = (1 - e^{-500\times10^{-6}t})$



At $t = t_0 = 500\,\mu sec$, S is changed to position ②



i_{LO}

$i_f = \frac{50}{100} = 0.5\,A$

$i_i = i(t)\big|_{t=500\mu sec}$

$i(500\times10^{-6}) = 1 - e^{-500\times500\times10^{-6}} = 1 - e^{-0.25}$

$= 1 - 0.7788$

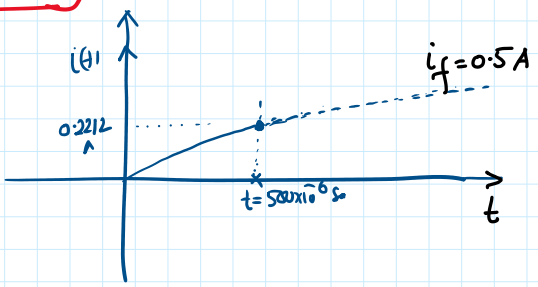
$i_{LO} = 0.2212\ A$

$i(t) = i_f + (i_i - i_f)\, e^{-500(t-t_0)}$

$= 0.5 + (0.2212 - 0.5)\, e^{-500(t-500\times10^{-6})}$

$i(t) = 0.5 - 0.2782\, e^{-500(t-500\times10^{-6})}$

At $t = t_0 = 500\,\mu sec$, $i(500\,\mu s) = 0.2212\,A$



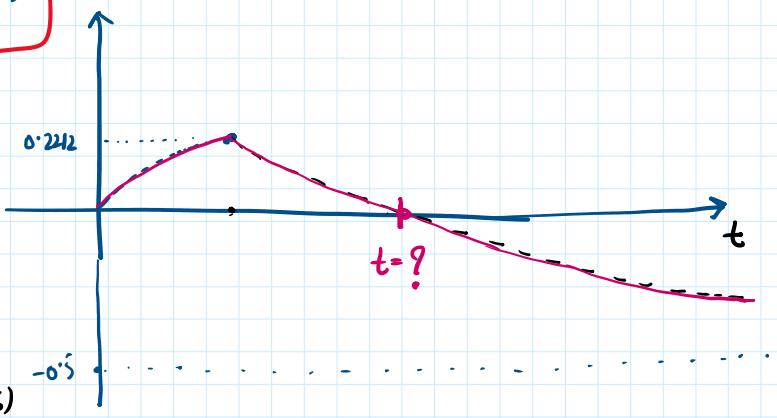
(iii) When polarity of 50V is reversed, $i_f = \frac{-50}{100} = -0.5\,A$

(iii) $i(t) = i_f + (i_i - i_f)\, e^{-\frac{R}{L}(t-t_0)}$

$= -0.5 + \{0.2212 - (-0.5)\}\, e^{-500(t-500\times10^{-6})}$

$i(t) = -0.5 + 0.7212\, e^{-500(t-500\times10^{-6})}$

$i_f = \frac{-E}{R} = \frac{-50}{100} = -0.5$



$t = t_1$, at which $i(t_1) = 0$

$0 = -0.5 + 0.7212\, e^{-500(t-500\times10^{-6})}$

$\Rightarrow e^{-500(t-500\times10^{-6})} = \frac{0.5}{0.7212}$

$-500(t-500\times10^{-6}) = \ln(0.6936)$

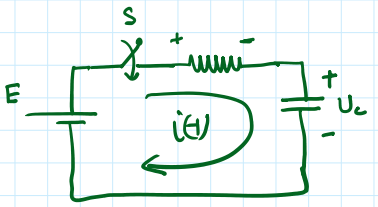
$t = 500\times10^{-6} + \frac{0.3664}{500}$

$= \left(500 + \frac{3664}{5}\right)\times10^{-6}$

$= (500 + 732.87)\times10^{-6}$

$t = 1232.87\,\mu s$

RC- Series Circuit



$E = iR + V_c$

$RC\frac{dV_c}{dt} + V_c = E$

$$\boxed{\frac{dV_c}{dt} + \frac{1}{RC}V_c = \frac{E}{RC}}$$

$V_c = \frac{1}{C}\int_c i\,dt \Rightarrow \boxed{C\frac{dV_c}{dt} = i}$

Solution of this first order differential equation is

$V_c(t) = Ke^{-at} + \frac{b}{a}$; $a = \frac{1}{RC}$; $b = \frac{E}{RC}$

$\frac{b}{a} = \frac{E/RC}{1/RC} = E$

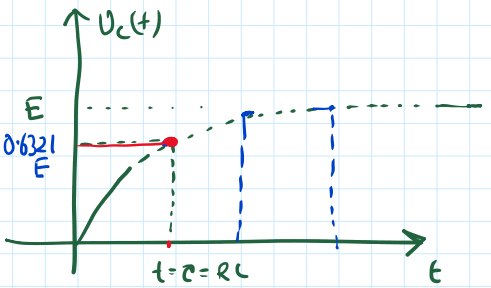
$$\boxed{V_c(t) = Ke^{-\frac{t}{RC}} + E}$$

K is solved using initial condition of voltage across the capacitor, i.e. say $V_c(0) = 0$

$\Rightarrow K = -E$. Substituting in main equation

$V_c(t) = E(1 - e^{-t/RC})$

$i_c = i = C\frac{dV_c}{dt} = \frac{E}{R}e^{-t/RC}$ —(**)



	t = 0	t → ∞
V_c	0	E
i_c	E/R	0
Nature	Short	Open

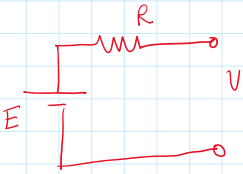
$V_c(t) = E(1 - e^{-t/\tau})$

τ = Time constant of RC circuit = RC



t = 0

Generalized Expression: $V_c = V_{co}$ at $t = t_o$



$V_c = E$ at $t \to \infty$

$V_c(t) = Ke^{-t/RC} + E$

$V_c(t_o) = V_{co} = Ke^{-t_o/RC} + E$

$K = (V_{co} - E)e^{t_o/RC}$

$V_c(t) = (V_{co} - E)e^{-(t-t_o)/RC} + E$

$= (V_{co} - E)e^{-(t-t_o)/RC}$

$$V_C(t) = (V_{C0} - E)\, e^{-(t-t_0)/RC} + E$$

$$V_C(t) = V_{C0}\, e^{-(t-t_0)/RC} + E\left\{1 - e^{-(t-t_0)/RC}\right\}$$

At $t = t_0$, $V_C(t_0) = V_{C0}$

$V_i = V_{C0}$, $V_f = E$

$$V_C(t) = V_i\, e^{-(t-t_0)/RC} + V_f\left\{1 - e^{-(t-t_0)/RC}\right\}$$

$$= V_f + (V_i - V_f)\, e^{-(t-t_0)/RC}$$

$$= V_i + (V_f - V_i)\left\{1 - e^{-(t-t_0)/RC}\right\}$$

An interesting mnemonic And: Suppose today you are student in the class at $t = t_0$. In future ($t \to \infty$), you want to become CEO of a company. Thus your journey is a move from student to CEO.

Intial point = Student

Final point = CEO

Difference = (CEO − Student) has to be brought to the final point, say through exponential rise i.e. $(1 - e^{-t/\tau_s})$, where

τ_s = Time constant of student, on how he responds. Then

$$CEO(t) = \text{Student} + (CEO - \text{student})\underbrace{\left(1 - e^{-t/\tau_s}\right)}_{\text{Exponential rise}}$$

In case of inductance, CEO $= i_{Li}$

student $= i_{Lf}$

hence

$$\boxed{i_L(t) = i_{Li} + (i_{Lf} - i_{Li})\left(1 - e^{-\frac{R}{L}t}\right)}$$

$$e^{-\frac{R}{L}t}$$

$$= i_{Li} + (i_{Lf} - i_{Li}) - (i_{Lf} - i_{Li}) e^{-\frac{R}{L}t}$$

$$\boxed{i_L(t) = i_{Lf} + (i_{Li} - i_{Lf}) e^{-R/L\,t}}$$

In case of capacitance; CEO = V_{cf}
Student = V_{ci}

$$\boxed{v_c(t) = v_{ci} + (v_{cf} - v_{ci})(1 - e^{-t/RC})}$$

$$= v_{ci} + (v_{cf} - v_{ci}) - (v_{cf} - v_{ci}) e^{-t/RC}$$

$$\boxed{v_c(t) = v_{cf} + (v_{ci} - v_{cf}) e^{-t/RC}}$$

$$\boxed{v_c(t) = v_{ci}\, e^{-t/RC} + v_{cf}(1 - e^{-t/RC})}$$